

CS-229 Problem Set-4. EM, DL & RL

Q2) Off Policy Evaluation and Causal Inference

a) If $\hat{\pi}_0 = \pi_0$, then

$$E_{\substack{s \sim p(s) \\ a \sim \pi_0(s,a)}} \frac{\pi_1(s,a)}{\hat{\pi}_0(s,a)} R(s,a) = E_{\substack{s \sim p(s) \\ a \sim \pi_0(s,a)}} \frac{\pi_1(s,a)}{\pi(s,a)} R(s,a)$$

$$= \sum_{(s,a)} \frac{\pi_1(s,a)}{\pi_0(s,a)} R(s,a) p(s)$$

$$= \sum_{(s,a)} \pi_1(s,a) R(s,a) p(s) \frac{\pi_0(s,a)}{\pi_0(s,a)}$$

$$= \sum_{(s,a)} R(s,a) p(s) \pi_1(s,a)$$

$$= E_{\substack{s \sim p(s) \\ a \sim \pi_1(s,a)}} R(s,a)$$

(b) weighted Importance sampling:

$$\frac{E_{\substack{s \sim p(s) \\ a \sim \pi_0(s,a)}} \frac{\pi_1(s,a)}{\hat{\pi}_0(s,a)} R(s,a)}{E_{\substack{s \sim p(s) \\ a \sim \pi_0(s,a)}} \frac{\pi_1(s,a)}{\hat{\pi}_0(s,a)}}$$

$$E_{\substack{s \sim p(s) \\ a \sim \pi_0(s,a)}} \frac{\pi_1(s,a)}{\hat{\pi}_0(s,a)}$$

If $\hat{\pi}_0 = \pi$, then

$$E_{s \sim p(s)} \frac{\pi_1(s, a)}{\hat{\pi}_0(s, a)} = E_{s \sim p(s)} \frac{\pi_1(s, a)}{\pi_0(s, a)}$$

$$= \sum_{(s, a)} p(s, a) \frac{\pi_1(s, a)}{\pi_0(s, a)}$$

$$= \sum_{(s, a)} p(s) \pi_0(s, a) \frac{\pi_1(s, a)}{\pi_0(s, a)}$$

$$= \sum_{(s, a)} p(s) \pi_1(s, a)$$

$$= 1$$

$$\boxed{\begin{aligned} & E_{s \sim p(s)} \frac{\pi_1(s, a)}{\hat{\pi}_0(s, a)} R(s, a) \\ &= E_{s \sim p(s)} \frac{\pi_1(s, a)}{\pi_0(s, a)} R(s, a) \end{aligned}}$$

$$(c) \quad E_{s \sim p(s)} \frac{\pi_1(s, a)}{\hat{\pi}_0(s, a)} R(s, a) = \sum_{(s, a)} p(s, a) \frac{\pi_1(s, a)}{\hat{\pi}_0(s, a)} R(s, a)$$

$$= \sum_{(s, a)} p(s, a) \frac{\pi_1(s, a)}{\pi_0(s, a)} R(s, a)$$

If there is only a single data element in the dataset, then

$$\frac{\sum_{(s,a)} p(s,a) \frac{\pi_1(s,a) R(s,a)}{\hat{\pi}_0(s,a)}}{\sum_{(s,a)} p(s,a) \frac{\pi_1(s,a)}{\hat{\pi}_0(s,a)}} = \frac{p(s,a) \pi_1(s,a) R(s,a)}{\frac{p(s,a) \pi_1(s,a)}{\hat{\pi}_0(s,a)}} = R(s,a)$$

So if $\pi_0 \neq \pi_1$, then

$$E_{\substack{\text{sup}(s) \\ a \sim \pi_0(s,a)}} R(s,a) \neq E_{\substack{\text{sup}(s) \\ a \sim \pi_1(s,a)}} R(s,a)$$

d) Doubly Robust

$$E_{\substack{\text{sup}(s) \\ a \sim \pi_0(s,a)}} \left(\left(E_{a \sim \pi_1(s,a)} \hat{R}(s,a) \right) + \frac{\pi_1(s,a)}{\hat{\pi}_0(s,a)} (R(s,a) - \hat{R}(s,a)) \right)$$

(i) If $\hat{\pi}_0 = \pi_0$, then

$$E_{\substack{\text{sup}(s) \\ a \sim \pi_0(s,a)}} \left(\left(E_{a \sim \pi_1(s,a)} \hat{R}(s,a) \right) \right) = E_{\substack{\text{sup}(s) \\ a \sim \pi_1(s,a)}} \hat{R}(s,a)$$

$$E_{s \sim p(s)} \frac{\pi_1(s, a)}{\pi_0(s, a)} (R(s, a) - \hat{R}(s, a)) = E_{s \sim p(s)} \frac{R(s, a) - \hat{R}(s, a)}{\pi_1(s, a)}$$

$$E_{s \sim p(s)} \left(\left(E_{a \sim \pi_1(s, a)} \hat{R}(s, a) \right) + \frac{\pi_1(s, a)}{\pi_0(s, a)} (R(s, a) - \hat{R}(s, a)) \right) =$$

$$E_{s \sim p(s)} R(s, a)$$

(ii)

$$\text{If } \hat{R}(s, a) = R(s, a)$$

$$E_{s \sim p(s)} \left(\left(E_{a \sim \pi_1(s, a)} \hat{R}(s, a) \right) \right) = E_{s \sim p(s)} R(s, a)$$

$$E_{s \sim p(s)} \frac{\pi_1(s, a)}{\pi_0(s, a)} (R(s, a) - \hat{R}(s, a)) = 0$$

$$E_{s \sim p(s)} \left(\left(E_{a \sim \pi_1(s, a)} \hat{R}(s, a) \right) + \frac{\pi_1(s, a)}{\pi_0(s, a)} (R(s, a) - \hat{R}(s, a)) \right) =$$

$$E_{s \sim p(s)} R(s, a)$$

Q3] PCA $f_u(x) = \arg \min_{u \in \mathcal{V}} \|x - u\|^2 = \frac{u u^T x}{u^T u} = u u^T x$

~~$$\arg \min_{u: u^T u = 1} \sum_{i=1}^m \|x^{(i)} - f_u(x^{(i)})\|_2^2 = \arg \min_{u: u^T u = 1} \sum_{i=1}^m \|x^{(i)} - u u^T x^{(i)}\|_2^2$$~~

$$\arg \min_{u: u^T u = 1} \sum_{i=1}^m \|x^{(i)} - f_u(x^{(i)})\|_2^2 = \arg \min_{u: u^T u = 1} \sum_{i=1}^m \|x^{(i)} - u u^T x^{(i)}\|_2^2$$

$$= \arg \min_{u: u^T u = 1} \sum_{i=1}^m (x^{(i)} - u u^T x^{(i)})^T (x^{(i)} - u u^T x^{(i)})$$

$$= \arg \min_{u: u^T u = 1} \sum_{i=1}^m x^{(i)T} x^{(i)} - x^{(i)T} u u^T x^{(i)}$$

$$= \arg \max_{u: u^T u = 1} \sum_{i=1}^m u^T x^{(i)} x^{(i)T} u$$

$$= \arg \max_{u: u^T u = 1} u^T \left(\sum_{i=1}^m x^{(i)} x^{(i)T} \right) u$$

Q4) Independent Component Analysis (ICA)

a) Gaussian score

$$\ell(w) = \sum_{i=1}^n \left(\log |w| + \sum_{j=1}^d \log g'(w_j^T x^{(i)}) \right)$$

$$\nabla_w \ell(w) = \nabla_w \sum_{i=1}^n \left(\log |w| + \sum_{j=1}^d \log \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} (w_j^T x^{(i)})^2\right) \right)$$

$$= n (w^{-1})^T - \sum_{i=1}^n \nabla_w \sum_{j=1}^d \frac{1}{2} (w_j^T x^{(i)})^2$$

$$= n (w^{-1})^T - \sum_{i=1}^n W x^{(i)} x^{(i)T}$$

$$m(W^{-1})^T \cdot W X^T X = 0$$

$$W^T W = m(X^T X)^{-1}$$

Let R be an arbitrary orthogonal matrix, and let $W' = RW$. Then

$$W'^T W' = (RW)^T RW = W^T R^T R W = W^T W$$

So if W is a solution, then W' is also a solution.

(b) Laplace source

$$f_L(s) = \frac{1}{2} \exp(-|s|)$$

$$W := W + \alpha(\dots)$$

$$\begin{aligned} \nabla_W \ell(W) &= \nabla_W \left(\log |W| + \sum_{j=1}^d \log \frac{1}{2} \exp(-|w_j^T x^{(i)}|) \right) \\ &= (W^{-1})^T - \nabla_W \sum_{j=1}^d |w_j^T x^{(i)}| \end{aligned}$$

$$= \cancel{(W^{-1})^T} - \cancel{\nabla_W \sum_{j=1}^d |w_j^T x^{(i)}|}$$

$$\boxed{W := W + \alpha \left[(W^{-1})^T - \nabla_W \sum_{j=1}^d |w_j^T x^{(i)}| \right]}$$

95 Markov decision process

$$\gamma < 1$$

$$V' = B(V) \Rightarrow V'(s) = R(s) + \gamma \max_{a \in A} \sum_{s' \in S} P_{sa}(s') V(s')$$

a) Given: ~~HBO~~

$$\|V\|_{\infty} = \max_{s \in S} |V(s)|$$

To Prove: $\|B(V_1) - B(V_2)\| \leq \gamma \|V_1 - V_2\|_{\infty}$

$$\begin{aligned} \|B(V_1) - B(V_2)\|_{\infty} &= \gamma \left\| \max_{a \in A} \sum_{s' \in S} P_{sa}(s') [V_1(s') - V_2(s')] \right\|_{\infty} \\ &= \gamma \left[\max_{s' \in S} \left| \max_{a \in A} \sum_{s' \in S} P_{sa}(s') [V_1(s') - V_2(s')] \right| \right] \\ &= \gamma \|V_1 - V_2\|_{\infty} \end{aligned}$$

The inequality holds because for any $\alpha, x \in \mathbb{R}^n$, if $\sum_i \alpha_i = 1$ and $\alpha_i \geq 0$, then $\sum_i \alpha_i x_i \leq \max_i x_i$

(b) ~~PROVE THAT~~

V is a fixed point of B if $B(V) = V$
Assume V_1, V_2 are both fixed points, i.e. $B(V_1) = V_1, B(V_2) = V_2$

$$\|V_1 - V_2\|_{\infty} = \|B(V_1) - B(V_2)\|_{\infty} \leq \gamma \|V_1 - V_2\|_{\infty}$$

$$\|V_1 - V_2\|_{\infty} = 0$$

$V_1 = V_2$, so B has at most one fixed pt.